

TKR-2500V/U

TRUNKING 2.5 VHF/UHF TDMA DIGITAL REPEATER



TRUNKING 2.5 was born as an unique technology for KENWOOD to provide a DMR-based trunking solution for shared-use frequencies. Providing a solid answer that offers capacity, flexibility, and capability, the TKR-2500 repeater brings forward KENWOOD's rich history of reliability and performance to our extensive range of customers.

Providing the backbone of TRUNKING 2.5, the TKR-2500V/U repeaters exclusively work with the KENWOOD NX-3000/5000 subscribers. The companion KTI-2500 controller allows for full scale system management and integration, as well as redundancy.



Features

- 50 - 2 W @ 100% Duty (136-174 MHz)
- 40 - 2 W @ 100% Duty (470-520 MHz)
- 40 - 2 W @ 100% Duty (400-470 MHz)
- 1U Height / 19-inch 1.71-inch
- Built-in IP Network Adapter
- Multicast
- CW ID
- SNMP Monitoring
- FCC Type 1 / Type 2 monitoring
- RADIUS / LDAP

Digital – Trunking 2.5

Hibernation / Quick CCH Rotation	Mixed	IP OTAP
Composite CCH		Priority Interrupt
Preference Level (CCH Priority)		Priority Call Notification
USBD (up to 1,000 per min)		Registration Timer
Authentication		Encryption (DES)
Text Messaging (Up to 46byte)		

Optional Features

KTI-2500 System Controller	G.711 Transcoder	Voting
AIS Calling	High Availability	
SIP Calling		



TDMA Trunking

Up to 16 talk paths per site, efficiently utilizing spectrum for high-capacity communication, dynamic channel allocation, and unmatched scalability.



High-Speed GPS

Real-time location tracking with pinpoint accuracy, ensuring optimal efficiency and safety.



DES Encryption

Robust digital encryption technology providing secure, confidential communication to safeguard sensitive information.

Specifications

General	TKR-2500VM	TKR-2500UM	TKR-2500UM2
Frequency Range	136 - 174 MHz	470 - 520 MHz	400-470 MHz
Channel Capacity	4000		
Channel Spacing Analog Digital	25 / 12.5* kHz 12.5 kHz		
PLL Channel Step	2.5 / 3.125 kHz	5 / 6.25 kHz	
Power Supply	12 - 16V DC		
Current Drain Standby Transmitting	0.66 A 9.6 A (50 W), 2.9 A (2 W)	0.7 A 8.1 A (40 W), 2.8 A (2 W)	
Operating Temperature	-22 °F to + 140 °F (-30 °C to +60 °C)		
Frequency Stability	± 0.5 ppm		
Antenna Impedance	50 Ω		
Weight Radio	14.6 lb (6.6 kg)		
FCC ID	CASTBDB1G	CASTBDH3G	CASTBDH5G

*25 kHz in VHF/UHF Bands excluding T-Band are not included in the models sold in the USA or US territories. Analog measurements made per TIA603. Specifications are measured according to applicable standards. Specifications shown are typical and subject to change without notice, due to advancements in technology. Details and timing of firmware and software updates are subject to change without notice.

Receiver	TKR-2500VM	TKR-2500UM	TKR-2500UM2
Sensitivity DMR (5 % BER) DMR (3 % BER) Analog (12dB SINAD)	0.18 µV 0.25 µV 0.25 µV		
Selectivity Analog @ 25 kHz (TIA603) Analog @ 25 kHz (TIA603D)	85 dB 50 dB		
FM Hum & Noise Analog @ 12.5kHz	45 dB		
Intermodulation	85 dB		
Spurious Rejection	100 dB		

Transmitter	TKR-2500VM	TKR-2500UM	TKR-2500UM2
RF Power Output	50 - 2 W @ 100% Duty	40 - 2 W @ 100% Duty	
Spurious Emission	9 kHz to 1 GHz 1 GHz to 4 GHz	30 MHz to 1 GHz 1 GHz to 4 GHz @ TX Freq. < 470 MHz 1 GHz to 12.75 GHz @ TX Freq. > 470 MHz	
FM Hum & Noise Analog @ 25kHz	55 dB		
Audio Distortion	2%		
Emission Designator	11K0F3E, 7K60FXD, 7K60FXW		